

Linux System Administration using Red Hat and Ubuntu

This course is focused on:

⇒ System Administration

Course Duration:

- **40 Hours**, 20 Classes, 2 Hours Per Session (Online)
- **42 Hours**, 14 Classes, 3 Hours per session (Offline)

Mod 01: Introduction to Linux Operating System Module 06: Linux Text Editors

- Introduction to Operating Systems
- Parts of Operating System
- Kernel, Shell & File
- History of UNIX and LINUX
- Linux and GNU Project
- Basic Concepts of Linux
- Identification of various Linux distributors

- Why need text editor
- Different types of text editors
- Introduction to 'vi/vim' and 'gedit'
- Linux Text Editor Utilities (vim, nano, gedit)
- Working with Different 'vi/vim' Modes
- Editing, Replacing, Searching with 'vi/vim'
- Working with '**vim**' advanced features

Mod 02: Preparing Linux Installation & Lab Setup

- Planning a Linux Installation
- Linux Home Lab Setup Requirements
- Partitioning Requirements
- Linux Installation Method (Physical, Virtual, Cloud)
- Required Partitions for RHEL 10 Installation
- Required Partitions for Ubuntu Installation
- Dual Booting with Windows Operating System
- Download and Install VMware Workstation

Module 07: User and Group Administration

- Users and Groups Introduction
- Linux User Types and Database
- Primary Groups and Supplementary Groups
- Gaining Super user Access
- Running commands as root with SUDO
- Managing Local User Accounts
- Managing Local Group Accounts
- Managing User Passwords
- Managing User's Password Aging

Mod 03: Linux Installations & Basic Configuration

- Introduction of Virtualization Technology
- Create VM on VMware Workstation for Linux OS
- Installation of RHEL on VMware Workstation
- Installation of Ubuntu on VMware Workstation
- Install RHEL & Ubuntu instance of AWS cloud
- Configure BIOS/UEFI options for OS booting
- Details discussion about OS booting options
- Details discussion about Installation Summary
- Linux Installation Method (MBR and GPT)
- Configure Post installation on RHEL 10 & Ubuntu

Module 08: Linux File Permissions and ACLs

- Explore Linux File & Directory Types
- Linux standard file permissions
- Hard Link and Soft Link concepts
- Viewing File/Directory Permission and Ownership
- Linux User, Group and Other permission Concept
- Set permission using read, write and execute
- Linux Special Permissions SUID, SGID, Sticky bit
- Securing Files with ACLs
- Creating, modifying and deleting ACL's

Module 04: Getting started with Linux

- The GNOME Desktop Environment
- Introduction to Linux shells and terminal
- Linux Virtual Console/Terminal
- Logging remote system through SSH
- Logging web interface using cockpit
- Linux Command Syntax, Options, Argument
- Examples of Simple Commands
- Linux Directory & File System introduction
- Navigating Linux Directory Paths
- Command-line File & Directory Management
- Files & Directory handling commands

Module 09: Linux Boot & Service Management

- Step by step Linux booting procedures
- Explain and Controlling the Boot Process
- Working with GRUB2 Boot loader
- Working with Linux Kernel
- Update Linux Kernel
- Introducing Linux Systemd
- Linux Process Management
- Details explain of "**TOP**" command
- Controlling Linux Daemon & Services
- Working with systemd unit and target
- Resetting root password

Module 05: Linux Text Processing Tools

- Standard Input, Output and Error Concept
- Redirecting Output to a File
- Constructing and Using Pipelines
- Working with tail, head, cat, less, wc, echo
- Working with Regular Expressions 'grep'
- Familiar with Linux 'find', ' ' and 'locate'
- Documentation for Commands

Module 10: Linux File System Management

- Understanding Linux file systems
- Identifying File Systems and Devices
- Managing MBR Partitions with '**fdisk**'
- Managing GPT Partitions with '**gdisk**'
- Creating File System (xfs, ext4, swap)
- Mount Points and '**/etc/fstab**' - Details
- Mounting and Un-mounting File Systems
- Working with USB, DVD, ISO, NTFS Partitions

Mod 11: Linux Network Management

- Understand Network Device Recognition
- Working with **NetworkManager** Services
- Introducing Network Manager tools (nmcli & nmtui)
- Configuring Host Name and Name Resolution
- Working with gateway and route table
- Configure Static and dynamic IP
- Configuring IPv4 Networking using '**nmtui**'
- Configuring IPv4 using GUI Interface

Mod 12: Linux Package Management System

- Software Packages and RPM
- The Linux Package Management system
- Examining RPM Package Files
- RPM Install, Queries and verifying
- Dependency problems and Resolution
- Concept of Package Repositories
- Register system with Red Hat Portal
- Packages Install and Remove using DNF
- Enable Third-party Software Repositories (EPEL)

Module 13: Configuring OpenSSH Service

- What is the Secure Shell (SSH)?
- How SSH (Secure Shell) works?
- SSH Host Keys (Public and Private)
- Configuring SSH Key-based Authentication
- Password less SSH Login
- Customizing SSH Service Configuration
- Restricting SSH Logins (root)
- Putty and Open SSH Clients

Mod 14: Linux Compression & Backup Utilities

- Why Need Backup?
- Different types of Backup method
- Archive Files and Directories using 'tar'
- Discuss about different '**tar**' options
- Extract Compressed '**tar**' Archive
- Discuss about **gz**, **bz2** and **xz** Compression
- Compress and De-compress using 'gz, bz2, xz'
- Secure Copy/Backup Through '**scp**'
- Working with Rsync file Backup

Module 15: Cron Process and Job Scheduling

- Introduction to Linux Scheduling
- Scheduling Recurring Jobs with 'cron'
- Scheduling System cron Jobs
- Explain cron job file format
- Running commands at particular times
- Identify various CRON entries and schedules
- Scheduling Future Linux Tasks

Module 16: Linux Security & Firewall Services

- Introduction to Firewall Technologies
- Netfilter and FirewallD concepts
- Working with '**firewalld**' zones concepts
- Managing & Configure Firewall Settings
- IP, ICMP, Port, Service Filtering using '**firewalld**'
- Introducing SELinux Security
- Explanation of SELinux Modes
- Introducing Ubuntu Firewall
- Implementing firewall filter service using UFW and FirewallD

